

2

:





1  
2  
3  
4  
5



6

7

8

( )



5in

13 7+13 8+14 2+14 3



1  $\Omega \subseteq \mathbb{R}^n$   $f: \Omega \rightarrow \mathbb{R}$   
1  $f \Omega$   $f \Omega$   
 $\Omega$   $n$   $\frac{\partial f}{\partial x_1} \dots \frac{\partial f}{\partial x_n}$   
 $f \Omega$   
2  $\frac{\partial f}{\partial x_1} \dots \frac{\partial f}{\partial x_n} X_0 \in \Omega$   $f$   
 $X_0$   
3  $f \Omega$   $f \Omega$   $\mathcal{C}^{(1)}(\Omega)$   
:



1.  $\Omega \subseteq \mathbb{R}^n$  ,  $f: \Omega \rightarrow \mathbb{R}$  ,  $X_0 \in \Omega$  ,  $f$   
:  
**Idea:** :  $n = 2$  :  
 $(h_1, h_2) \rightarrow (0,0)$

$$\frac{\partial_{x_1}(x_1^{(0)}x_2^{(0)})h_1 + \frac{\partial f}{\partial x_2}(x_1^{(0)}x_2^{(0)})h_2}{\mathcal{O}(\|(h_1, h_2)\|)}$$

$$f(x_1^{(0)} + h_1x_2^{(0)} + h_2) - f(x_1^{(0)}x_2^{(0)})$$



$$\begin{aligned}
& \because n=2 \quad f \\
& X_0 \quad \exists r>0 \quad f \\
& B(X_0\sqrt{2}r) \quad X_0 \\
& X_0=(x_1^{(0)}\,x_2^{(0)})\,\forall h_1h_2\in(-rr) \\
\\
& =\bigl(\mathbf{f}(\mathbf{x}_1^{(0)}+h_1x_2^{(0)}+h_2)-\textcolor{red}{f}(x_1^{(0)}\,x_2^{(0)}+\textcolor{red}{h_2})\bigr) \\
& +\bigl(\mathbf{f}(\textcolor{red}{\mathbf{x}}_1^{(0)}\,\textcolor{red}{x_2^{(0)}}+\textcolor{red}{h_2})-f(x_1^{(0)}\,x_2^{(0)})\bigr)
\end{aligned}$$

$$F(h_1h_2)=f(x_1^{(0)}+h_1x_2^{(0)}+h_2)-f(x_1^{(0)}\,x_2^{(0)})$$



Lagrange      $\exists \theta_1 \theta_2 \in (01)$

$$+\partial f_{\partial x_2(x_1^{(0)}x_2^{(0)}+\theta_2h_2)h_2}$$

$$F(h_1h_2)=\frac{\partial f}{\partial x_1}(x_1^{(0)}+\theta_1h_1x_2^{(0)}+h_2)h_1$$

$$\lim_{(h_1h_2)\rightarrow(00)}\theta_1h_1=\lim_{(h_1h_2)\rightarrow(00)}\theta_2h_2=0$$



$$f\;X_0$$

$$\lim_{(h_1h_2)\rightarrow(00)}\frac{\partial f}{\partial x_2}(x_1^{(0)}x_2^{(0)}+\textcolor{red}{\theta_2}h_2)=\frac{\partial f}{\partial x_2}(x_1^{(0)}x_2^{(0)})\quad (h_1h_2)\rightarrow(00)$$

$$+\overline{\left(\frac{\partial f}{\partial x_2(x_1^{(0)}x_2^{(0)})+o(1)}\right)h_2}$$

$$\lim_{(h_1h_2)\rightarrow(00)}\frac{\partial f}{\partial x_1}(x_1^{(0)}+\textcolor{red}{\theta_1}h_1x_2^{(0)}+h_2)=\frac{\partial f}{\partial x_1}(x_1^{(0)}x_2^{(0)})$$

$$F(h_1h_2)=\left(\frac{\partial f}{\partial x_1}(x_1^{(0)}x_2^{(0)})+o(1)\right)h_1$$



$$|h_1| \leq \sqrt{h_1^2 + h_2^2} \; |h_2| \leq \sqrt{h_1^2 + h_2^2} \quad (h_1 h_2) \rightarrow (00)$$

$$\begin{array}{l} 5\mathbf{cm} + \mathbf{o(1)h_1} + o(1)h_2 \\ \hline \frac{\partial f}{\partial x_1(x_1^{(0)}\;x_2^{(0)})h_1 + \frac{\partial f}{\partial x_2}(x_1^{(0)}\;x_2^{(0)})h_2 + o(1)\sqrt{h_1^2 + h_2^2}} \\ \hline \frac{\partial f}{\partial x_1(x_1^{(0)}\;x_2^{(0)})h_1 + \frac{\partial f}{\partial x_2}(x_1^{(0)}\;x_2^{(0)})h_2 + o(\|(h_1h_2)\|)} \end{array}$$

$$f\;X_0$$

$$F(h_1h_2)=\frac{\partial f}{\partial x_1}(x_1^{(0)}\;x_2^{(0)})h_1+\frac{\partial f}{\partial x_2}(x_1^{(0)}\;x_2^{(0)})h_2$$



$$\begin{aligned} &\textcolor{red}{1} \quad \forall (xyz) \in \mathbb{R}^3 \setminus \{(000)\} \\ f(xyz) &= x^2 + e^{xyz} + \ln(x^2 + y^2 + z^2) \\ &\int\limits_{f\mathbb{R}^3 \setminus \{(000)\}} \\ &\quad f\mathbb{R}^3 \setminus \{(000)\} \end{aligned}$$

$$\begin{aligned} &+\left(\mathbf{xze}^{xyz}+\frac{2y}{x^2+y^2+z^2}\right)\mathrm{d}y \\ &+\left(\mathbf{xye}^{xyz}+\frac{2z}{x^2+y^2+z^2}\right)\mathrm{d}z \end{aligned}$$

$$\mathrm{d}f(xyz)=\left(2x+yz e^{xyz}+\frac{2x}{x^2+y^2+z^2}\right)\mathrm{d}x$$



$$\forall (xyz) \in \mathbb{R}^3$$

$$f(0,0)$$

$$\frac{\partial f}{\partial y}(0,0)=0 \quad (x,y)\rightarrow (0,0)$$

$$f(0,0)=\mathrm{d}f(0,0)=0$$

$$f(xy)=\begin{cases} (x^2+y^2)\sin\frac{1}{\sqrt{x^2+y^2}} & (xy)\neq (0,0) \\ 0 & (xy)=(0,0) \end{cases}$$

$$\frac{\partial f}{\partial x}(0,0)=\lim_{h\rightarrow 0}\frac{f(h,0)-f(0,0)}{h}=\lim_{h\rightarrow 0}\frac{h^2\sin\frac{1}{|h|}}{h}=0$$

$$|f(xy)-f(0,0)|=(x^2+y^2)|\sin\frac{1}{\sqrt{x^2+y^2}}|=o(\|(xy)\|)$$



$$(xy) \neq (00)$$

$$\frac{\partial f}{\partial y}(xy)=2y\sin\frac{1}{\sqrt{x^2+y^2}}-\frac{y}{\sqrt{x^2+y^2}}\cos\frac{1}{\sqrt{x^2+y^2}}\qquad \forall \textcolor{red}{k}\in\mathbb{R}$$

$$\lim_{x\rightarrow 0}\frac{\partial f}{\partial x}(xkx) \qquad \frac{\partial f}{\partial x} \quad (00)$$

$$\frac{\partial f}{\partial x}(xy)=2x\sin\frac{1}{\sqrt{x^2+y^2}}-\frac{x}{\sqrt{x^2+y^2}}\cos\frac{1}{\sqrt{x^2+y^2}}$$

$$\frac{\partial f}{\partial x}(\textcolor{red}{x}\textcolor{red}{k}\textcolor{red}{x})=2x\sin\frac{1}{\sqrt{1+\textcolor{red}{k}^2}|x|}-\frac{\operatorname{sgn}x}{\sqrt{1+\textcolor{red}{k}^2}}\cos\frac{1}{\sqrt{1+\textcolor{red}{k}^2}|x|}$$



3  $f: \mathbb{R}^2 \rightarrow \mathbb{R}$   $\mathbb{R}^2 : f \mathbb{R}^2$   
 $\exists M > 0 \quad \forall (xy) \in \mathbb{R}^2$   
 $|\frac{\partial f}{\partial y}(xy)| \leq M \quad (x_0y_0) \in \mathbb{R}^2 \quad \forall (xy) \in \mathbb{R}^2$   
Lagrange  $\xi \quad y_0y$

$=|f(x_0y_0) - f(x_0y_0)| + |\frac{\partial f}{\partial y}(x\xi)||y - y_0|$   
 $\leq |f(x_0y_0) - f(x_0y_0)| + M|y - y_0|$

$|f(xy) - f(x_0y_0)| \leq |f(xy_0) - f(x_0y_0)| + |f(xy) - f(xy_0)|$







5in

14 1



$$\textcolor{blue}{1} \quad \Omega \subseteq \mathbb{R}^n \quad X_0 \in \Omega \, f \colon \Omega \rightarrow \mathbb{R} \quad \beta \neq 0 \quad \beta^0 = (\cos \alpha_1 \cdots \cos \alpha_n)^T \quad \alpha_j \, \beta \, x_j \quad :$$

$$\lim_{\textcolor{red}{h} \rightarrow 0^+} \frac{f(X_0 + h \beta^0) - f(X_0)}{\textcolor{red}{h}} \quad f \mid_{X_0 + \beta} \quad \frac{\partial f}{\partial \beta}(X_0) \, \frac{\partial f}{\partial \beta} \big|_{X_0} \, f_{\beta}'(X_0) \quad \frac{\partial f}{\partial \beta}(X_0) = \frac{\partial f}{\partial \beta^0}(X_0)$$

$$\beta^0 = \frac{\beta}{\|\beta\|} = (\cos \alpha_1 \cdots \cos \alpha_n)^T$$



2

$$= (a_1\,a_2\,a_n); \quad B = (b_1\,b_2\,b_n);$$

$$\cdot B = a_1\,b_1 + a_2\,b_2 + \; + a_n\,b_n$$

2in  $f\colon \Omega \rightarrow \mathbb{R} \Omega \subset \mathbb{R}^n$

$$\nabla f(X_0) := \left(\frac{\partial f}{\partial x_1}(X_0)\frac{\partial f}{\partial x_2}(X_0)\frac{\partial f}{\partial x_n}(X_0)\right)$$



$$\mathbf{1} \quad f(X_0) \quad \frac{\partial f}{\partial \beta}(X_0)$$

$$= \nabla f(X_0) \cdot \beta^0 = \nabla f(X_0) \cdot \frac{\beta}{\|\beta\|} \quad : \quad f(X_0) \quad h \rightarrow 0^+$$

$$= \left(\sum_{j=1}^n \frac{\partial f}{\partial x_j}(X_0) \cos \alpha_j\right) h + o(|h|)$$

$$\frac{\partial f}{\partial \beta}(X_0) = \sum_{j=1}^n \frac{\partial f}{\partial x_j}(X_0) \cos \alpha_j$$

$$f(X_0 + \beta^0 h) - f(X_0) = \sum_{j=1}^n \frac{\partial f}{\partial x_j}(X_0) h \cos \alpha_j + o(\|\beta^0 h\|)$$



$$\begin{aligned}
& \textcolor{blue}{1} \quad \forall (xy) \in \mathbb{R}^2 \quad f(xy) = 1 - \sqrt{x^2 + y^2} \\
& : \quad f \text{ (00)} \\
& : \quad \lim_{h \rightarrow 0} \frac{f(h0) - f(00)}{h} = \lim_{h \rightarrow 0} \frac{-|h|}{h} \\
& f \\
& \beta^0 = (\cos \alpha \sin \alpha)^T \\
& \frac{\partial f}{\partial \beta^0}(00) = \lim_{h \rightarrow 0^+} \frac{f(h \cos \alpha \sin \alpha) - f(00)}{h} = -1
\end{aligned}$$







1

$\Omega \subseteq \mathbb{R}^n$ 
 $\Omega$

2

$\Omega \subseteq \mathbb{R}^n$ 
 $X_0 \in \Omega$ 
 $f: \Omega \rightarrow \mathbb{R}$

$f(X_0)$ 
 $f(X_0)$

$\text{grad} f(X_0)$ 
 $\nabla f(X_0)$ 
 $\overrightarrow{\text{grad} f(X_0)}$

$\mathbf{grad} f(X_0)$ 
 $\nabla f(X_0)$



3  $\mathbb{R}^n$

$$X \cdot Y = \sum_{j=1}^n x_j y_j \quad XY$$

$$X = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \quad Y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \in \mathbb{R}^n$$



1  $f(X_0)$

$$\text{grad}f(X_0) = \begin{pmatrix} \frac{\partial f}{\partial x_1}(X_0) \\ \vdots \\ \frac{\partial f}{\partial x_n}(X_0) \end{pmatrix}$$



$$\textcolor{blue}{:} \qquad \beta^0 = (\cos \alpha_1 \cdots \cos \alpha_n)^T$$

$$=\check{\mathbf{e}}\cdot\beta^0=|||\cos\langle\beta^0\rangle\quad\langle\beta^0\rangle\quad\beta^0$$

$$\frac{\partial f}{\partial \beta^0}(X_0) = \sum_{j=1}^n \frac{\partial f}{\partial x_j}(X_0) \cos \alpha_j$$



$\frac{\partial f}{\partial \beta^0}(X_0)$

$$\frac{\partial f}{\partial \beta^0}(X_0) = \text{grad}f(X_0) \cdot \beta^0$$



$uv = ab$

- $\text{grad} c = 0 \quad c$
- $\text{grad}(au + bv) = a\text{grad} u + b\text{grad} v$
- $\text{grad}(uv) = v\text{grad} u + u\text{grad} v$
- $\text{grad}\left(\frac{u}{v}\right) = \frac{1}{v^2}(v\text{grad} u - u\text{grad} v) \quad v \neq 0$
- $\text{grad} f(u) = f'(u)\text{grad} u(f)$



$$\mathbf{1} \quad \forall (xyz) \in \mathbb{R}^3 \setminus \{(000)\}$$

$$\begin{array}{l} f \quad P_0 = (100) \\ \qquad \beta = (12-2)^T \\ : \end{array}$$

$$f(xyz) = x^2 + e^{xyz} + \log(x^2 + y^2 + z^2)$$

$$\operatorname{grad} f(xyz) = \begin{pmatrix} 2x + yze^{xyz} + \frac{2x}{x^2+y^2+z^2} \\ xze^{xyz} + \frac{2y}{x^2+y^2+z^2} \\ xye^{xyz} + \frac{2z}{x^2+y^2+z^2} \end{pmatrix}$$



$$\begin{aligned} f(P_0) - \text{grad}f(P_0) &= \begin{pmatrix} 4 \\ 0 \\ 0 \end{pmatrix} \\ P_0 \quad \|\text{grad}f(P_0)\| &= 4 \\ \beta &= \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix} \quad \beta^0 = \begin{pmatrix} \frac{1}{3} \\ \frac{2}{3} \\ -\frac{2}{3} \end{pmatrix} \quad f(P_0) - \beta \end{aligned}$$

$$\frac{\partial f}{\partial \beta}(P_0) = \text{grad}f(P_0) \cdot \beta^0 = \frac{4}{3}$$



$$\begin{aligned}
& \textcolor{blue}{2} \quad \frac{f(x_0y_0)}{\beta_1^0\beta_2^0} = 3dx - 2dy \\
& : \quad \frac{\beta_1^0}{\beta_1^0} \frac{\beta_2^0}{\beta_2^0} \frac{f(x_0y_0)}{\beta_1^0} = -\beta_1^0 \quad \frac{\beta_2^0}{\partial x(x_0y_0)} = 3 \quad \frac{\partial f}{\partial y}(x_0y_0) = -2 \\
& \beta_1^0 = \begin{pmatrix} \frac{3\sqrt{13}}{13} \\ -\frac{2\sqrt{13}}{13} \end{pmatrix} \quad \beta_2^0 = \begin{pmatrix} -\frac{3\sqrt{13}}{13} \\ \frac{2\sqrt{13}}{13} \end{pmatrix}
\end{aligned}$$



5in

14.9



$$\begin{aligned}
 & \text{1 } \Omega \subset \mathbb{R}^n \qquad f: \Omega \rightarrow \mathbb{R} \\
 & \frac{\partial f}{\partial x_i} \quad (1 \leq i \leq n) \qquad \frac{\partial f}{\partial x_i} \quad j \\
 & \frac{\partial}{\partial x_j} \left( \frac{\partial f}{\partial x_i} \right) \\
 & \frac{\partial^2 f}{\partial x_j \partial x_i} \quad \partial_{ji} f \qquad i = j \\
 & \frac{\partial^2 f}{\partial x_i^2}
 \end{aligned}$$



⋮  
⋮  
1

?

$\forall (xy) \in \mathbb{R}^2$

$\forall (xy) \in \mathbb{R}^2 \quad y \neq 0 \quad f(xy) = \frac{(x^2-y^2)xy}{x^2+y^2}$

$f(xy) = xy - \frac{2xy^3}{x^2+y^2}$

$$f(xy) = \begin{cases} \frac{(x^2-y^2)xy}{x^2+y^2} & (xy) \neq (00) \\ 0 & (xy) = (00) \end{cases}$$



$$\begin{aligned}
 &= y - 2y^3 \frac{1}{x^2 + y^2 + \frac{2xy^3}{(x^2 + y^2)^2}} (2x) \\
 &= (x^2 - y^2) y \frac{1}{x^2 + y^2 + \frac{4x^2 y^3}{(x^2 + y^2)^2}}
 \end{aligned}$$

$$\frac{\partial f}{\partial x}(0,y) = -y$$

$$\frac{\partial f}{\partial x}(xy) = \frac{\partial}{\partial x}\Big(xy - \frac{2xy^3}{x^2 + y^2}\Big)$$



$$\forall y \in \mathbb{R}$$

$$\frac{\partial f}{\partial x}(00) = \lim_{x \rightarrow 0} \frac{f(x0) - f(00)}{x} = 0$$

$$\frac{\partial f}{\partial x}(0y) = -y$$

$$\frac{\partial f}{\partial y}(x0) = x$$



$$\frac{\partial^2 f}{\partial x \partial y}(00) = \lim_{x \rightarrow 0} \frac{\frac{\partial f}{\partial y}(x0) - \frac{\partial f}{\partial y}(00)}{x} = 1$$

∴ ?

$$\frac{\partial^2 f}{\partial y \partial x}(00) = \lim_{y \rightarrow 0} \frac{\frac{\partial f}{\partial x}(0y) - \frac{\partial f}{\partial x}(00)}{y} = -1$$



$$\begin{aligned} 1 \quad & \Omega \subset \mathbb{R}^n \quad f: \Omega \rightarrow \mathbb{R} \\ \Omega \quad & \frac{\partial^2 f}{\partial x_j \partial x_i} = \frac{\partial^2 f}{\partial x_i \partial x_j} \quad (i \neq j) \\ & X_0 \in \Omega \end{aligned}$$

$$: \qquad n=2$$

$$\frac{\partial^2 f}{\textcolor{red}{\partial x_j} \textcolor{blue}{\partial x_i}}(X_0) = \frac{\partial^2 f}{\textcolor{blue}{\partial x_i} \textcolor{red}{\partial x_j}}(X_0)$$

$$\frac{\partial f}{\partial x_2}(x_1x_2) = \lim_{h_2 \rightarrow 0} \frac{f(x_1x_2+h_2) - f(x_1x_2)}{h_2}$$



$$X_0 = (a_1\,a_2)$$

$$\begin{aligned} &= \lim_{h_1 \rightarrow 0} \frac{1}{h_1} \Big( \lim_{h_2 \rightarrow 0} \frac{f(a_1+h_1\,a_2+h_2)-f(a_1+h_1\,a_2)}{h_2} \\ &\quad - \lim_{h_2 \rightarrow 0} \frac{f(a_1\,a_2+h_2)-f(a_1\,a_2)}{h_2} \Big) \\ &= \lim_{h_1 \rightarrow 0} \lim_{h_2 \rightarrow 0} \frac{1}{h_1 h_2} \Big( (f(a_1+h_1\,a_2+h_2)-f(a_1+h_1\,a_2)) \\ &\quad - (f(\mathbf{a}_1\,a_2+h_2)-f(a_1\,a_2)) \Big) \quad \mathbf{1} \end{aligned}$$

$$\frac{\partial^2 f}{\partial x_1 \partial x_2}(a_1\,a_2) = \lim_{h_1 \rightarrow 0} \frac{\frac{\partial f}{\partial x_2}(a_1+h_1\,a_2) - \frac{\partial f}{\partial x_2}(a_1\,a_2)}{h_1}$$







$$\varphi(x_1) = f(x_1 a_2 + h_2) - f(x_1 a_2)$$

$$\frac{\partial f}{\partial x_1} \quad \varphi \qquad \textbf{Lagrange} \qquad \exists \theta_1 \in (01)$$

$$= \left( \frac{\partial f}{\partial x_1(a_1 + \theta_1 h_1 a_2 + h_2)} \right. \\ \left. - \frac{\partial f}{\partial x_1(a_1 + \theta_1 h_1 a_2)} \right) h_1$$

$$F(h_1 h_2) = \varphi(a_1 + h_1) - \varphi(a_1)$$

$$F(h_1 h_2) = \varphi'(a_1 + \theta_1 h_1) h_1$$



$$\frac{\partial^2 f}{\partial x_2 \partial x_1} \qquad \text{Lagrange} \qquad \exists \theta_2 \in (0,1)$$

$$= \frac{\partial}{\partial x_2 \left( \frac{\partial f}{\partial x_1} \right) (a_1 + \theta_1 h_1 a_2 + \theta_2 h_2) h_1 h_2}$$

$$= \frac{\partial^2 f}{\partial x_2 \partial x_1 (a_1 a_2)}$$

$$F(h_1 h_2) = \left( \frac{\partial f}{\partial x_1} (a_1 + \theta_1 h_1 a_2 + h_2) - \frac{\partial f}{\partial x_1} (a_1 + \theta_1 h_1 a_2) \right) h_1$$

$$\lim_{(h_1 h_2) \rightarrow (0,0)} \frac{F(h_1 h_2)}{h_1 h_2} = \lim_{(h_1 h_2) \rightarrow (0,0)} \frac{\partial^2 f}{\partial x_2 \partial x_1} (a_1 + \theta_1 h_1 a_2 + \theta_2 h_2)$$



$$\begin{aligned} & \mathbb{R}^n \subset \mathbb{R}^n \quad k \geq 0 \\ & \mathcal{C}^{(k)}(\Omega) = \{f \mid f \in \mathcal{C}^k(\Omega) \} \\ & \mathcal{C}^{(0)}(\Omega) = \mathcal{C}(\Omega) \quad f \in \mathcal{C}^{(k)}(\Omega) \\ & f \in \mathcal{C}^k(\Omega) \quad k \geq 2 \quad f \in \mathcal{C}^{(k)}(\Omega) \\ & 1 \leq r \leq k \quad f \in \mathcal{C}^{(r)}(\Omega) \end{aligned}$$



